

Measuring time

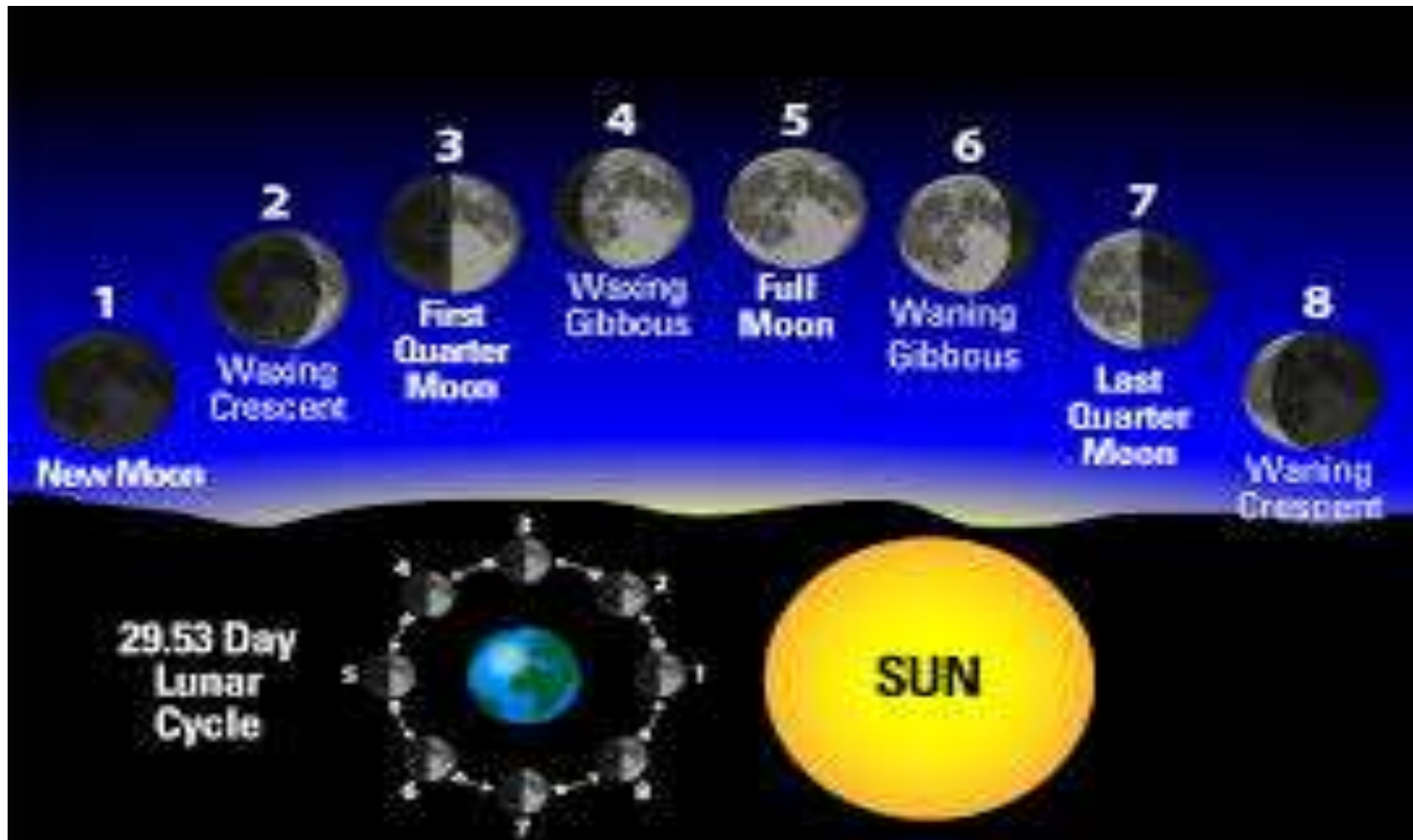
- Time can be measured by any moving object which goes on repeating its motion after a fixed interval of time. Eg sun, moon, swinging pendulum.
- The SI unit of time is second.
- The rising and setting of the sun and moon has a relatively fixed period. Allah Ta'ala states;

We made the night and the day as two signs. We made the moon non luminous, and the sun luminous, so that you (all) can seek your lord's fadal, and you can know the computation of years and hisab. (read also Arabic text)

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- The moon's phases form the basis for making a lunar calendar.
- The *Misri* calendar that we use is calculated according to the Moon's journey in its stages.
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Cont'd : phases of the moon



Note

- Allah Ta'ala states

Indeed salat is compulsory upon all mumineen at (its) stated times. (read also Arabic text pg 10)

For *namaaz*, time is of crucial importance. In *kitab al-taharat*, the entire section is devoted just to show the calculations for *namaaz* timings. Similarly, timing is crucial for doing *rozah*

Measuring a *ghari*

Measurement of *ghari* help in calculating *namaz* timings

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- One *ghari* for a day is calculated by taking the total time between the ***rising and setting*** of the sun and dividing it by twelve.
- One *ghari* for a night is calculated by taking the total time between the ***setting and rising*** of the sun and dividing by twelve
- Suppose the amount of time between sunset on lailat al-qadar and sunrise the next morning is 624minites
- To calculate one *ghari* we divide this number by twelve. The resulting *ghari* would be

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- The movement of sun and moon helps us in measuring long intervals of time, like a year.
- For measuring small intervals of time we use clocks and watches
- Mechanical clocks are use repetitive oscillatory processes, like swing of a pendulum.
- For sprinters, a fraction of second (0.01sec) can make a difference between winning, second or third.
- Today we have atomic clocks which measures time with credible precision.
- Atomic clocks are most accurate time and frequency standards known

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- Are used as primary standards for international time distribution services, to control the wave frequency of television broadcasts and in global navigation satellite systems such as GPS
- It maintains an accuracy of 1.0×10^{-9} sec per day.

Model of an atomic clock



Theory of errors

- Errors in measurement are unavoidable they can be systematic, gross or random
- ***Systematic errors*** always either overestimate or underestimate the results of measurements and arise for specific reasons (incorrect set-up of measuring equipment, the effect of environment, etc.), which systematically affect the measurements and alter them in one direction
- ***Gross errors*** (often called outliers) arise from miscalculations, incorrect reading of the measuring equipment, etc

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- ***Random errors*** arise from various reasons which have an unforeseen effect on each of the measurements, both in overestimating and in underestimating results

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- The theory points to the fact that the only true perfection is in Allah Ta'ala' measurement
- He states;

And we will set a just balance for the day of judgment so that no soul faces injustice. And though it (a deed) will be of the weight of a grain of mustard seed, we will bring it forth, and we suffice for justice and evaluation (pg 11; read also Arabic text)

- Allah Ta'ala states

And Allah has counted all things

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- Allah Ta'ala knows the exact number of everything in the universe that can be its order, value, measurement, weight or any other property of a thing that you can assign a number to- it is all present in Allah Ta'ala's knowledge
- Accurate measurement is essential and therefore approximation is required which makes the measurement to be close to accurate
- Allah Ta'ala guides his followers to do proper measurement. He states

Give full measure and weigh with an accurate balance

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On the other hand Allah Ta'ala condemns those who cheat. He states;

Woe is to the cheaters! Those who take more when they weigh and give less

- Scientific experiments are carried out with great care to eliminate as much errors as possible
- Precision in measurement is very important in many fields like in Engineering.
- Al-Dai al-Ajal Syedna Mohammed Burhanuddin states in a *ziyafat* given by engineers; (*read Arabic text pg 12 to link the knowledge*)

Summary

- Allah Ta'ala has written the universe in the language of mathematics
- The fundamental quantities are length, mass and time, and their SI units are M, kg, and Sec respectively
- **Derived units** are dependent on the fundamental units while fundamental units are independent
- **Area** is the measure of the surface occupied by an object
- **Volume** is the measure of the space occupied by an object
- The **mass** of an object is the amount of matter it is made of.

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- **Density** is the property of a material and shows the concentration of mass in it
- **Time** can be measured by an object which has a periodic motion
- Scholars believe that accurate measurement is very difficult and therefore errors in measurement are unavoidable.

Extensions

Improving precision in measurement (LAB)

- Vernier Calipers
- Micrometer screw gauge